PRAKTIKUM SISTEM OPERASI

MODUL 3

EKSPLORASI XINU

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

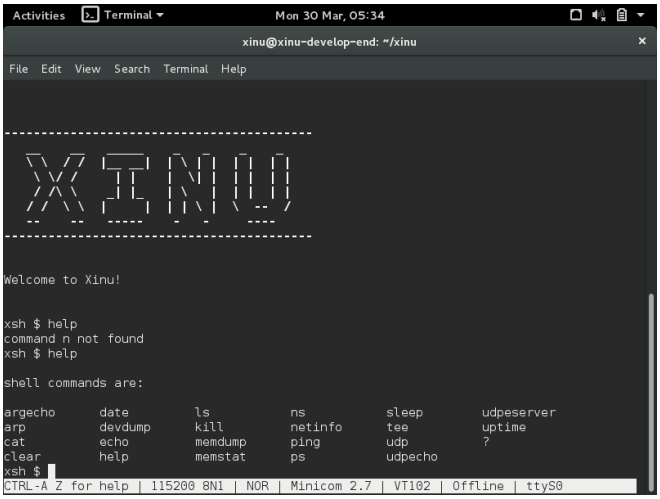
2026

JURNAL

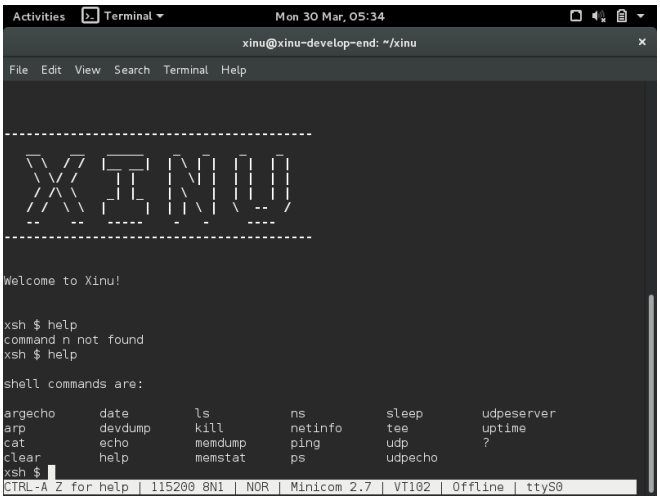
1. Jalankan Xinu OS, kemudian eksekusi perintah-perintah yang tersedia pada shell Xinu. Selanjutnya, uraikan serta jelaskan fungsi dari setiap perintah yang terdapat di dalam Xinu!

   Jawaban:

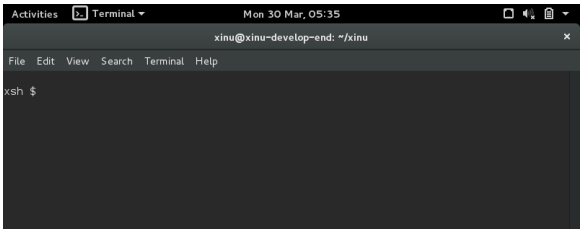
   a. help: untuk menampilkan semua perintah yang tersedia pada Xinu



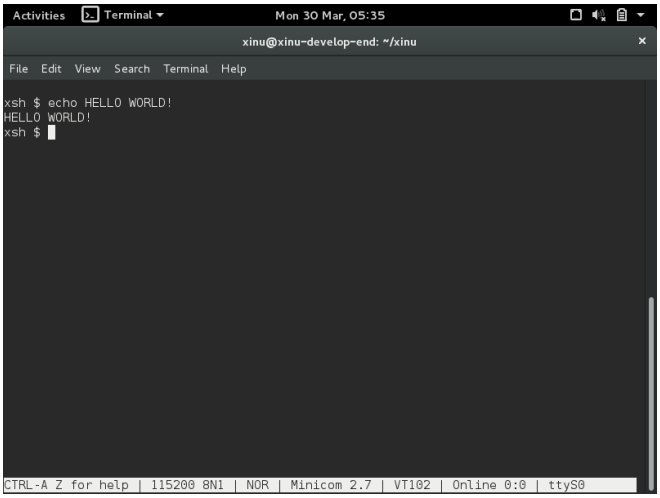
   b. ?: sama seperti help, yaitu menampilkan daftar perintah



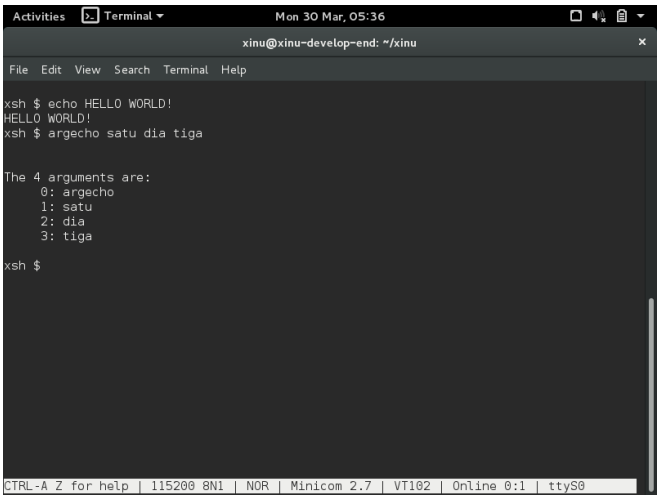
   c. clear: untuk membersihkan tampilan terminal



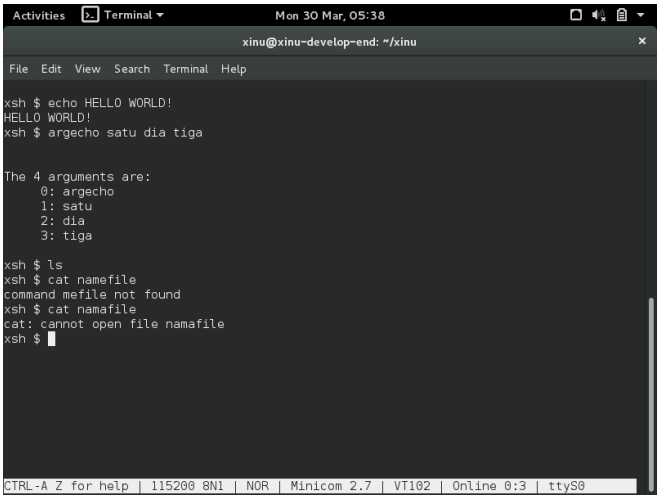
d. echo: untuk menampilkan teks atau string ke terminal



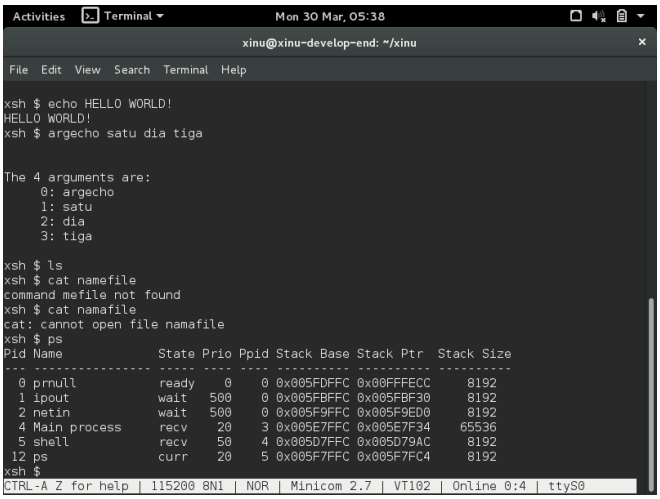
e. argecho: untuk menampilkan kembali argumen yang dimasukkan user



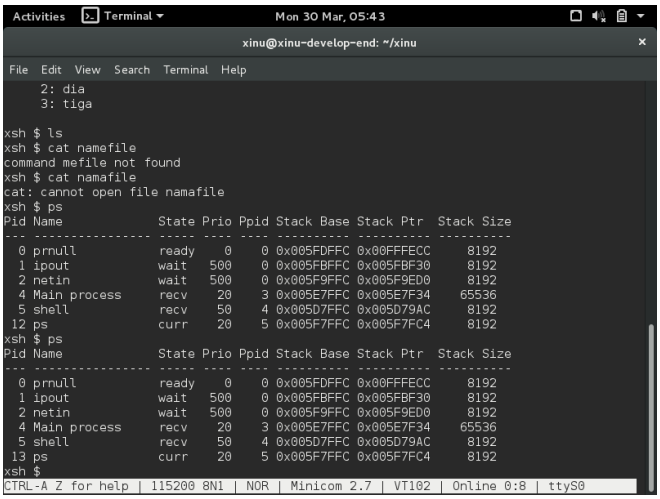
f. ls: untuk menampilkan daftar file dalam direktori



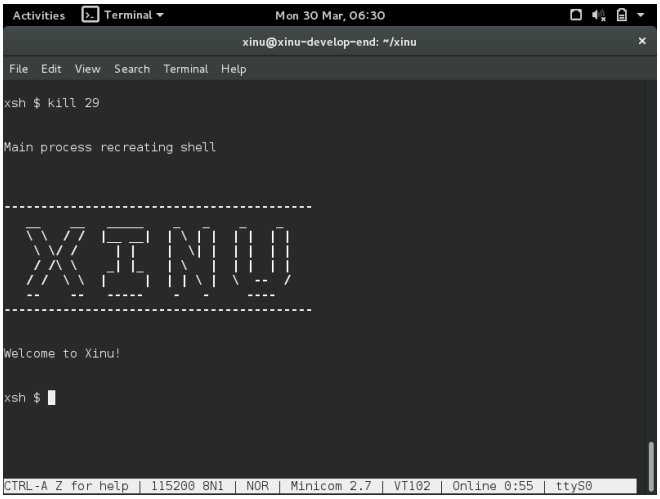
g. cat: untuk menampilkan isi file



h. ps: untuk menampilkan daftar proses yang sedang berjalan



i. kill: untuk menghentikan (terminasi) proses berdasarkan PID



j. date: untuk menampilkan tanggal dan waktu sistem

```
Activities  Terminal  Mon 30 Mar, 05:44
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Help
xsh $ ls
xsh $ cat namefile
command mefile not found
xsh $ cat namafile
cat: cannot open file namafile
xsh $ ps
Pid Name             State Prio Ppid Stack Base Stack Ptr  Stack Size
--- ---------------- ----- ---- ---- ---------- ---------- ----------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFECC     8192
  1 ipout            wait   500    0 0x005FBFFC 0x005FBF30     8192
  2 netin            wait   500    0 0x005F9FFC 0x005F9ED0     8192
  4 Main process     recv    20    3 0x005E7FFC 0x005E7F34    65536
  5 shell            recv    50    4 0x005D7FFC 0x005D79AC     8192
 12 ps               curr    20    5 0x005F7FFC 0x005F7FC4     8192
xsh $ ps
Pid Name             State Prio Ppid Stack Base Stack Ptr  Stack Size
--- ---------------- ----- ---- ---- ---------- ---------- ----------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFECC     8192
  1 ipout            wait   500    0 0x005FBFFC 0x005FBF30     8192
  2 netin            wait   500    0 0x005F9FFC 0x005F9ED0     8192
  4 Main process     recv    20    3 0x005E7FFC 0x005E7F34    65536
  5 shell            recv    50    4 0x005D7FFC 0x005D79AC     8192
 13 ps               curr    20    5 0x005F7FFC 0x005F7FC4     8192
xsh $ sleep 5
xsh $ date
Mon  Mar 30   8:43:35 EDT 2026
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:9 | ttyS0
```

k. sleep: untuk menunda proses selama beberapa detik

```
Activities  Terminal  Mon 30 Mar, 05:44
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Help
xsh $ ls
xsh $ cat namefile
command mefile not found
xsh $ cat namafile
cat: cannot open file namafile
xsh $ ps
Pid Name             State Prio Ppid Stack Base Stack Ptr  Stack Size
--- ---------------- ----- ---- ---- ---------- ---------- ----------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFECC     8192
  1 ipout            wait   500    0 0x005FBFFC 0x005FBF30     8192
  2 netin            wait   500    0 0x005F9FFC 0x005F9ED0     8192
  4 Main process     recv    20    3 0x005E7FFC 0x005E7F34    65536
  5 shell            recv    50    4 0x005D7FFC 0x005D79AC     8192
 12 ps               curr    20    5 0x005F7FFC 0x005F7FC4     8192
xsh $ ps
Pid Name             State Prio Ppid Stack Base Stack Ptr  Stack Size
--- ---------------- ----- ---- ---- ---------- ---------- ----------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFECC     8192
  1 ipout            wait   500    0 0x005FBFFC 0x005FBF30     8192
  2 netin            wait   500    0 0x005F9FFC 0x005F9ED0     8192
  4 Main process     recv    20    3 0x005E7FFC 0x005E7F34    65536
  5 shell            recv    50    4 0x005D7FFC 0x005D79AC     8192
 13 ps               curr    20    5 0x005F7FFC 0x005F7FC4     8192
xsh $ sleep 5
xsh $ date
Mon  Mar 30   8:43:35 EDT 2026
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:9 | ttyS0
```

l. uptime: untuk menampilkan lama waktu sistem telah berjalan

```
Activities  Terminal  Mon 30 Mar, 05:45
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Help
command mefile not found
xsh $ cat namafile
cat: cannot open file namafile
xsh $ ps
Pid Name             State Prio Ppid Stack Base Stack Ptr  Stack Size
--- ---------------- ----- ---- ---- ---------- ---------- ----------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFECC     8192
  1 ipout            wait   500    0 0x005FBFFC 0x005FBF30     8192
  2 netin            wait   500    0 0x005F9FFC 0x005F9ED0     8192
  4 Main process     recv    20    3 0x005E7FFC 0x005E7F34    65536
  5 shell            recv    50    4 0x005D7FFC 0x005D79AC     8192
 12 ps               curr    20    5 0x005F7FFC 0x005F7FC4     8192
xsh $ ps
Pid Name             State Prio Ppid Stack Base Stack Ptr  Stack Size
--- ---------------- ----- ---- ---- ---------- ---------- ----------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFECC     8192
  1 ipout            wait   500    0 0x005FBFFC 0x005FBF30     8192
  2 netin            wait   500    0 0x005F9FFC 0x005F9ED0     8192
  4 Main process     recv    20    3 0x005E7FFC 0x005E7F34    65536
  5 shell            recv    50    4 0x005D7FFC 0x005D79AC     8192
 13 ps               curr    20    5 0x005F7FFC 0x005F7FC4     8192
xsh $ sleep 5
xsh $ date
Mon  Mar 30   8:43:35 EDT 2026
xsh $ uptime
Xinu has been up  2 minute(s)  38 second(s)
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:10 | ttyS0
```

m. memstat: untuk menampilkan statistik penggunaan memori

```
Activities  Terminal  Mon 30 Mar, 05:45
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Help
--- --------------- ----- ---- ---- ---------- ---------- ----------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFECC     8192
  1 ipout            wait   500    0 0x005FBFFC 0x005FBF30     8192
  2 netin            wait   500    0 0x005F9FFC 0x005F9ED0     8192
  4 Main process     recv    20    3 0x005E7FFC 0x005E7F34    65536
  5 shell            recv    50    4 0x005D7FFC 0x005D79AC     8192
 13 ps               curr    20    5 0x005F7FFC 0x005F7FC4     8192
xsh $ sleep 5
xsh $ date
Mon  Mar 30   8:43:35 EDT 2026
xsh $ uptime
Xinu has been up  2 minute(s)  38 second(s)
xsh $ memstat
Current system memory statistics:
--------------------------------
    282563 bytes (0x00044fc3) of Xinu code
     98304 bytes (0x00018000) of allocated stack space
   4054760 bytes (0x003ddee8) of available kernel heap space

Free List:
Block address  Length (dec)  Length (hex)
-------------  ------------  ------------
  0x0018e110      -975128     0xfff11ee8
  0x00100000      5070848     0x004d6000
  0x005e8000        57344     0x0000e000

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:10 | ttyS0
```

n. memdump: untuk menampilkan isi memori

```
Activities  Terminal  Mon 30 Mar, 05:46
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Help
  2 netin            wait   500    0 0x005F9FFC 0x005F9ED0     8192
  4 Main process     recv    20    3 0x005E7FFC 0x005E7F34    65536
  5 shell            recv    50    4 0x005D7FFC 0x005D79AC     8192
 13 ps               curr    20    5 0x005F7FFC 0x005F7FC4     8192
xsh $ sleep 5
xsh $ date
Mon  Mar 30   8:43:35 EDT 2026
xsh $ uptime
Xinu has been up  2 minute(s)  38 second(s)
xsh $ memstat
Current system memory statistics:
--------------------------------
    282563 bytes (0x00044fc3) of Xinu code
     98304 bytes (0x00018000) of allocated stack space
   4054760 bytes (0x003ddee8) of available kernel heap space

Free List:
Block address  Length (dec)  Length (hex)
-------------  ------------  ------------
  0x0018e110      -975128     0xfff11ee8
  0x00100000      5070848     0x004d6000
  0x005e8000        57344     0x0000e000

xsh $ memdump
memdump: incorrect number of arguments
Try 'memdump --help' for more information
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:11 | ttyS0
```

o. devdump: untuk menampilkan informasi perangkat (device)

```
Activities  Terminal  Mon 30 Mar, 05:46
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Help
xsh $ memdump
memdump: incorrect number of arguments
Try 'memdump --help' for more information
xsh $ devdump
Device     Name     Minor
------ ------------ -----
   0   CONSOLE        0
   1   NULLDEV        0
   2   ETHER0         0
   3   NAMESPACE      0
   4   RDISK          0
   5   RAM0           0
   6   RFILESYS       0
   7   RFILE0         0
   8   RFILE1         1
   9   RFILE2         2
  10   RFILE3         3
  11   RFILE4         4
  12   RFILE5         5
  13   RFILE6         6
  14   RFILE7         7
  15   RFILE8         8
  16   RFILE9         9
  17   LFILESYS       0
  18   LFILE0         0
  19   LFILE1         1
  20   LFILE2         2
  21   LFILE3         3
```

```
Activities  Terminal  Mon 30 Mar, 05:46
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Help
  9   RFILE2         2
 10   RFILE3         3
 11   RFILE4         4
 12   RFILE5         5
 13   RFILE6         6
 14   RFILE7         7
 15   RFILE8         8
 16   RFILE9         9
 17   LFILESYS       0
 18   LFILE0         0
 19   LFILE1         1
 20   LFILE2         2
 21   LFILE3         3
 22   LFILE4         4
 23   LFILE5         5
 24   PIPE           0
 25   PIPE0          0
 26   PIPE1          1
 27   PIPE2          2
 28   PIPE3          3
 29   PIPE4          4
 30   PIPE5          5
 31   PIPE6          6
 32   PIPE7          7
 33   PIPE8          8
 34   PIPE9          9
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:12 | ttyS0
```

p. netinfo: untuk menampilkan informasi jaringan

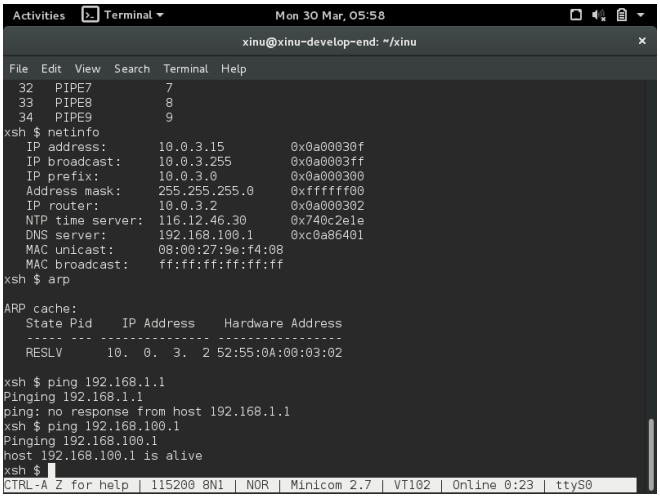
```
Activities  Terminal  Mon 30 Mar, 05:47
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Help
 19   LFILE1         1
 20   LFILE2         2
 21   LFILE3         3
 22   LFILE4         4
 23   LFILE5         5
 24   PIPE           0
 25   PIPE0          0
 26   PIPE1          1
 27   PIPE2          2
 28   PIPE3          3
 29   PIPE4          4
 30   PIPE5          5
 31   PIPE6          6
 32   PIPE7          7
 33   PIPE8          8
 34   PIPE9          9
xsh $ netinfo
   IP address:       10.0.3.15         0x0a00030f
   IP broadcast:     10.0.3.255        0x0a0003ff
   IP prefix:        10.0.3.0          0x0a000300
   Address mask:     255.255.255.0     0xffffff00
   IP router:        10.0.3.2          0x0a000302
   NTP time server:  116.12.46.30      0x740c2e1e
   DNS server:       192.168.100.1     0xc0a86401
   MAC unicast:      08:00:27:9e:f4:08
   MAC broadcast:    ff:ff:ff:ff:ff:ff
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:12 | ttyS0
```

q. arp: untuk menampilkan tabel ARP (relasi IP dengan MAC address)

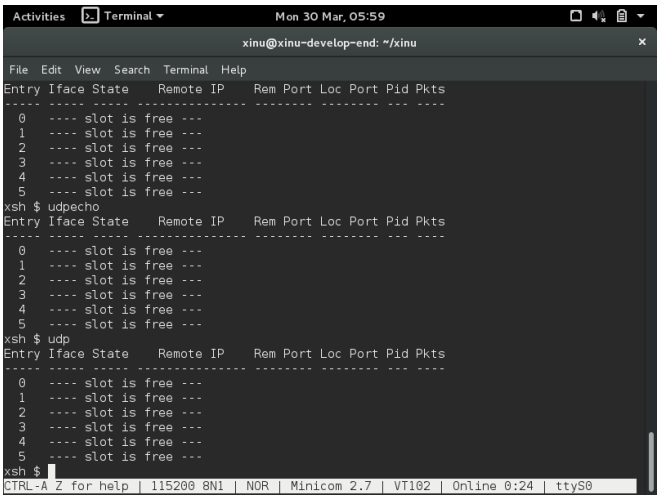
```
Activities  Terminal  Mon 30 Mar, 05:47
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Help
 26   PIPE1          1
 27   PIPE2          2
 28   PIPE3          3
 29   PIPE4          4
 30   PIPE5          5
 31   PIPE6          6
 32   PIPE7          7
 33   PIPE8          8
 34   PIPE9          9
xsh $ netinfo
   IP address:       10.0.3.15         0x0a00030f
   IP broadcast:     10.0.3.255        0x0a0003ff
   IP prefix:        10.0.3.0          0x0a000300
   Address mask:     255.255.255.0     0xffffff00
   IP router:        10.0.3.2          0x0a000302
   NTP time server:  116.12.46.30      0x740c2e1e
   DNS server:       192.168.100.1     0xc0a86401
   MAC unicast:      08:00:27:9e:f4:08
   MAC broadcast:    ff:ff:ff:ff:ff:ff
xsh $ arp

ARP cache:
   State Pid    IP Address    Hardware Address
   ----- --- --------------- -----------------
   RESLV      10.  0.  3.  2 52:55:0A:00:03:02

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:12 | ttyS0
```

e

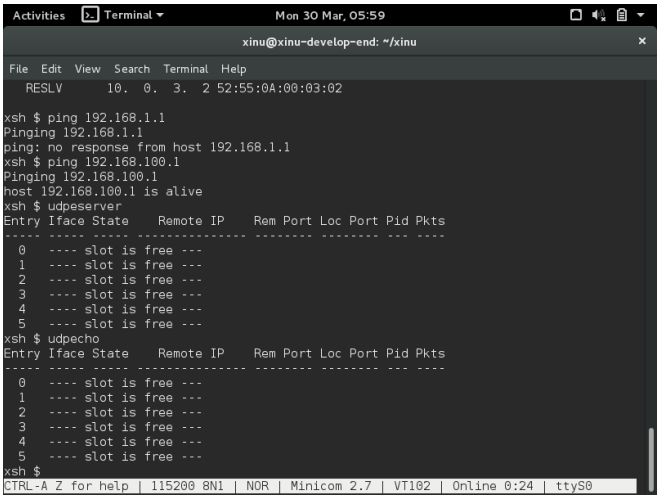
r. ping: untuk menguji koneksi ke suatu alamat IP



s. udp: untuk mengirim data menggunakan protokol UDP



t. udpecho: untuk mengirim dan menerima kembali pesan UDP



u. udpeserver: untuk menjalankan server UDP

```
Activities  Terminal  Mon 30 Mar, 05:58
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Help
   NTP time server:  116.12.46.30      0x740c2e1e
   DNS server:       192.168.100.1     0xc0a86401
   MAC unicast:      08:00:27:9e:f4:08
   MAC broadcast:    ff:ff:ff:ff:ff:ff
xsh $ arp

ARP cache:
   State Pid    IP Address    Hardware Address
   ----- --- --------------- -----------------
   RESLV      10.  0.  3.  2 52:55:0A:00:03:02

xsh $ ping 192.168.1.1
Pinging 192.168.1.1
ping: no response from host 192.168.1.1
xsh $ ping 192.168.100.1
Pinging 192.168.100.1
host 192.168.100.1 is alive
xsh $ udpeserver
Entry Iface State    Remote IP    Rem Port Loc Port Pid Pkts
----- ----- ----- --------------- -------- -------- --- ----
  0   ---- slot is free ---
  1   ---- slot is free ---
  2   ---- slot is free ---
  3   ---- slot is free ---
  4   ---- slot is free ---
  5   ---- slot is free ---
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:24 | ttyS0
```

v. tee: untuk menampilkan output sekaligus menyimpannya ke file

```
Activities  Terminal  Mon 30 Mar, 06:00
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Help
  0   ---- slot is free ---
  1   ---- slot is free ---
  2   ---- slot is free ---
  3   ---- slot is free ---
  4   ---- slot is free ---
  5   ---- slot is free ---
xsh $ udpecho
Entry Iface State    Remote IP    Rem Port Loc Port Pid Pkts
----- ----- ----- --------------- -------- -------- --- ----
  0   ---- slot is free ---
  1   ---- slot is free ---
  2   ---- slot is free ---
  3   ---- slot is free ---
  4   ---- slot is free ---
  5   ---- slot is free ---
xsh $ udp
Entry Iface State    Remote IP    Rem Port Loc Port Pid Pkts
----- ----- ----- --------------- -------- -------- --- ----
  0   ---- slot is free ---
  1   ---- slot is free ---
  2   ---- slot is free ---
  3   ---- slot is free ---
  4   ---- slot is free ---
  5   ---- slot is free ---
xsh $ echo TEST | tee file.txt
TEST
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:25 | ttyS0
```

w. untuk melakukan query ke DNS server

```
xsh $ ns google.com
 google.com becomes google.com on device 17
xsh $
```

x. exit: untuk keluar dari shell Xinu

```
xsh $ exit
Shell closed


Main process recreating shell



-----------------------------------------
   __   __  _____   _  _   _  _
   \ \  / /  |__ __|  | \ | |  | |  | |
    \ \/ /     | |    |  \| |  | |  | |
    / /\ \    _| |_   | \   |  | |  | |
   / /  \ \  |       | | \ |  \  --  /
   --    --  -----    -   -    ----
-----------------------------------------


Welcome to Xinu!


xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:26 | ttyS0
```

2. Jawablah pertanyaan-pertanyaan berikut ini:
   a. Berapaa jumlah perintah pada Xinu?
      Jawaban: 23 perintah

   b. Sebutkan 2 perintah yang mempunyai fungsi yang sama!
      Jawaban: help dan ? (sama-sama menampilkan daftar perintah)

   c. Berapa IP address Xinu?
      Jawaban: 10.0.3.15

   d. Perintah apa yang digunakan untuk mengetahui IP address?
      Jawaban: netinfo

   e. Berapa IP DNS server yang digunakan untuk mengetahui IP address?
      Jawaban: 192.168.100.1

   f. Terdapat berapa proses yang sedang digunakan oleh Xinu?
      Jawaban: 5 proses

```
xsh $ ps
Pid Name             State Prio Ppid Stack Base Stack Ptr  Stack Size
--- ---------------- ----- ---- ---- ---------- ---------- ----------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFECC     8192
  1 ipout            wait   500    0 0x005FBFFC 0x005FBF30     8192
  2 netin            wait   500    0 0x005F9FFC 0x005F9ED0     8192
  4 Main process     recv    20    3 0x005E7FFC 0x005E7F34    65536
 29 shell            recv    20    4 0x005F7FFC 0x005F79AC     4096
 30 ps               curr    20   29 0x005F6FFC 0x005F6DFC     8192
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:42 | ttyS0
```

   g. Proses apa yang mempunyai prioritas paling rendah?
      Jawaban: prnull (prioritas = 0, paling kecil)

   h. Proses apa yang mempunyai ukuran paling besar?
      Jawaban: Main process (Stack Size = 65536)

   i. Proses apa yang berada dalam state current?
      Jawaban: Main process

   j. Proses apa yang berada dalam state suspend?
      Jawaban: Tidak ada proses dalam state suspend
      (yang ada: ready, wait, curr, recv)

   k. Berapa PID (Process ID) dari Main process?
      Jawaban: 4

l. Berapa PPID (Parent Process ID) dari Main process?
   Jawaban: 0

m. Terminasi proses ps!
   Jawaban:

```
xsh $ ps
Pid Name             State Prio Ppid Stack Base Stack Ptr  Stack Size
--- ---------------- ----- ---- ---- ---------- ---------- ----------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFECC     8192
  1 ipout            wait   500    0 0x005FBFFC 0x005FBF30     8192
  2 netin            wait   500    0 0x005F9FFC 0x005F9ED0     8192
  4 Main process     recv    20    3 0x005E7FFC 0x005E7F34    65536
 29 shell            recv    20    4 0x005F7FFC 0x005F79AC     4096
 30 ps               curr    20   29 0x005F6FFC 0x005F6DFC     8192
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:42 | ttyS0
```

n. Terminasi proses shell!
   Jawaban:

```
Activities  Terminal  Mon 30 Mar, 06:30
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Help

xsh $ kill 29

Main process recreating shell

----------------------------------------
  __   __   ____   _  _   _   _
  \ \  / /  |_ _|  | \ | |  | |  | |
   \ \/ /    | |   |  \| |  | |  | |
   / /\ \    _| |_  | \  |  | |  | |
  / /  \ \  |    |  | | \ |  \ --  /
  --   --  -----   -  -    ----
----------------------------------------

Welcome to Xinu!

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:55 | ttyS0
```

o. Berapa lama Xinu sudah berjalan?
   Jawaban:

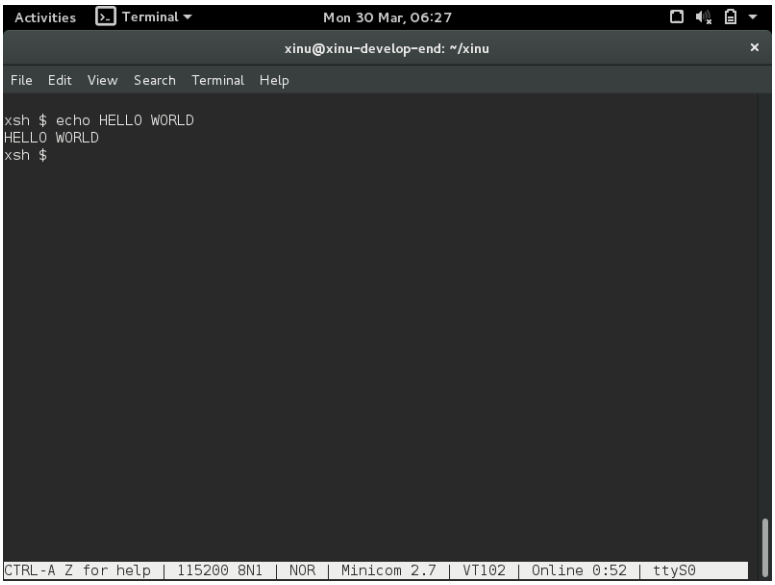
```
xsh $ uptime
Xinu has been up  2 minute(s)  38 second(s)
```

p. Perintah apa yang digunakan untuk membersihkan tampilan terminal?
   Jawaban: clear

q. Perintah apa yang digunakan untuk menampilkan statistik memory pada Xinu?
   Jawaban: memstat

r. Lakukan ping ke DNS server!
   Jawaban: ping 192.168.100.1

s. Lakukan perintah untuk menampilkan “HELLO WORLD!” pada terminal!
   Jawaban: echo HELLO WORLD



t. Silahkan keluar dari terminal Xinu!
   Jawaban: exit